

Revision — 1.3.1 Compression, Encryption and Hashing

OCR H446 — one-page revision summary

Must know

- **Compression** reduces file size → less storage, less bandwidth, faster transfer/download.
- **Lossless**: no data permanently removed; original is exactly recoverable (text, code, ZIP, PNG).
Lossy: data permanently removed; original **cannot** be recovered but files are much smaller (JPEG, MP3, video).
- **Run Length Encoding (RLE)**: lossless; replaces *runs* of repeated values with one value + a count. Good for long runs (simple images); can *enlarge* files with few/short runs.
- **Dictionary compression**: lossless; stores recurring patterns once in a dictionary/index and replaces each with a short reference. The dictionary must be sent with the file (good for text, e.g. LZW).
- **Encryption** scrambles plaintext → ciphertext using an algorithm + key, protecting confidentiality if data is intercepted.
- **Symmetric**: same key encrypts and decrypts; fast, but has the **key-distribution problem**.
Asymmetric: public + private key pair; slower, but the public key can be shared openly (solves distribution).
- Asymmetric: data encrypted with the **public** key can only be decrypted by the matching **private** key.
Digital signature: sign with private key, verify with public key (proves origin/integrity). HTTPS uses asymmetric to exchange a symmetric session key, then symmetric for bulk data.
- **Hashing**: a one-way function producing a fixed-length **hash/digest**. Properties: one-way (not reversible), deterministic, few collisions, fast. Used for storing passwords (store the hash) and hash tables ($\approx O(1)$ lookup).
- **Hashing $\neq$ encryption**: hashing is one-way and has no key; encryption is reversible with a key. Passwords are **hashed**, not encrypted.

Key terms

Term	Definition
Lossless compression	Reduces size with no permanent data loss; original exactly recoverable
Lossy compression	Permanently removes data to shrink files; original not recoverable
Run Length Encoding	Lossless; replaces runs of repeated values with value + count
Dictionary compression	Lossless; stores repeated patterns once and references them
Symmetric encryption	Same single key used to encrypt and decrypt
Asymmetric encryption	Public key encrypts, matching private key decrypts
Digital signature	Data signed with private key, verified with public key to prove origin
Hashing	One-way function producing a fixed-length digest from an input

Worked example — RLE

Encode the pixel run W W W W W B B W W W W W W (5 W, 2 B, 6 W):

Encoded: 5W 2B 6W → 3 value+count pairs instead of 13 values.

Decode 5A 3B 1C 4A back to the original: AAAAA + BBB + C + AAAA = AAAAABBBCAAAA .

Note: if the run were W B W B W B , RLE gives 1W 1B 1W 1B 1W 1B — **larger** than the original, showing RLE only helps with long runs.

Exam tips

- "Lossy = reduced quality" is not the same as "recoverable" — lossy data **cannot** be restored. State this explicitly.
- Always say RLE does **not** always shrink a file; it only helps when there are long runs of repeated values.
- For hashing questions, stress it is **one-way** and used for **passwords** (store the hash, compare hashes at login) — do not call it encryption.
- For HTTPS/secure transfer, mention asymmetric is used to share a symmetric key, then symmetric encrypts the bulk data (asymmetric alone is too slow).